

Achievers — Level 2 — Answer Key

1. **C) 225** — Original = $360 \div 1.20 = 300$. Decreasing 300 by 25%: $300 \times 0.75 = 225$.
2. **D) 125** — Total parts = $2+3+5 = 10$. 1 part = $250 \div 10 = 25$. Largest = $5 \times 25 = 125$.
3. **C) ₹4,200** — After spending $2/7$ on rent, $5/7$ of the money remains. After spending 15% of that on groceries, 85% of $5/7 = 17/28$ of the original remains. So $17/28$ of the total = ₹2,550 $\rightarrow$ total = ₹4,200.
4. **C) 10** — Let the daughter's age be x ; mother's age = $3x$. After 10 years: $3x+10 = 2(x+10) \rightarrow 3x+10 = 2x+20 \rightarrow x = 10$.
5. **B) 6 hours** — Pipe A fills $1/10$ of the tank per hour, Pipe B fills $1/15$ per hour. Together: $1/10 + 1/15 = 1/6$ of the tank per hour, so the tank fills in 6 hours.
6. **C) 33** — Let the numbers be $x, x+2, x+4, x+6$. Sum: $4x+12 = 120 \rightarrow x = 27$. Largest = $27+6 = 33$.
7. **C) 80** — Larger number = (sum + difference) $\div 2 = (134+26) \div 2 = 80$.
8. **C) 12** — Let a = number of ₹10 coins, b = number of ₹5 coins. $a+b = 22$ and $10a+5b = 170$. Substituting $b = 22-a$: $10a + 5(22-a) = 170 \rightarrow 5a + 110 = 170 \rightarrow a = 12$.
9. **C) 7 h 30 min** — Speed = $300 \div 5 = 60$ km/h. Time for 450 km = $450 \div 60 = 7.5$ hours = 7 h 30 min.
10. **B) 24** — Let the tens digit be t ; the ones digit is $t+2$. The number = $11t+2$, and the digit sum = $2t+2$. Setting $11t+2 = 4(2t+2)$ gives $11t+2 = 8t+8 \rightarrow t = 2$. Number = $11 \times 2 + 2 = 24$. Check: digit sum = $2+4 = 6$, and $4 \times 6 = 24$. ✓
11. **C) 425 sq cm** — Perimeter = $2(l+b) = 84 \rightarrow l+b = 42$. Since $l = b+8$: $2b+8 = 42 \rightarrow b = 17, l = 25$. Area = $17 \times 25 = 425$ sq cm.
12. **D) 158** — Each term is double the previous term plus 2: $38 \times 2 + 2 = 78, 78 \times 2 + 2 = 158$.
13. **C) ₹1,710** — After a 10% discount: $₹2,000 \times 0.90 = ₹1,800$. After a further 5% discount: $₹1,800 \times 0.95 = ₹1,710$.
14. **C) ₹5,000** — The interest earned between year 4 and year 6 is $₹8,000 - ₹7,000 = ₹1,000$ over 2 years, so ₹500 per year. Principal = amount at year 4 minus 4 years' interest = $₹7,000 - (4 \times ₹500) = ₹5,000$.
15. **C) 123** — The number must leave remainder 4 when divided by 7 and remainder 3 when divided by 5. Checking numbers between 100 and 150 that fit both conditions gives 123 ($123 \div 7 = 17$ remainder 4; $123 \div 5 = 24$ remainder 3).

How to use this: 10+/15 is strong for a Level-2-focused set. Q1, Q13, and Q14 are the ones most students underrate — successive-percentage-change and two-point simple-interest problems come up

constantly at Level 2, so if any of these tripped you up, redo similar ones from Week 5 first before retrying this set.